Online Resource 1 for

An exact line-integral formulation for evaluating tesseroid potentials and their derivatives through third order

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Introduction

Online Resource 1 provides the graphical and statistical evidence supporting the main article. It presents relative-error, absolute-error, and direct-value comparisons for polar-axis Tests No. 2–5; the two additional radial configurations, Tests No. 7 and 8, of the complete-spherical-shell experiment; and the below-source and in-source Laplace–Poisson Tests No. 10 and 11. Tables S1–S11 are generated directly from the archived pointwise outputs by the statistical audit scripts in the accompanying `Analysis` directory; they report the count, minimum, maximum, mean, RMSE, P50, P90, and P99 without suppressing near-zero-reference cases, while the audit log reports nonfinite counts separately. Tables S12–S13 are independently generated from all 160,801 timing records for Test No. 12 and document both the componentwise distributions and derivative-rank costs. The complete mathematical derivation and implementation details are supplied separately in Online Resource 2 (`ESM_2.pdf`).

The relative discrepancies in Test No. 1 use the larger magnitude of the two compared, radius-normalized values as the denominator, with the recorded numerical floor of 10^{-100}. The polar-axis relative errors use the analytical reference magnitude, as defined below. For Tests No. 2–11, $G\rho = 1\ \mathrm{s}^{-2}$ and the length unit is kilometres; an order-m potential derivative and its absolute error have units $\mathrm{km}^{2-m}\,\mathrm{s}^{-2}$. Logarithmic absolute-error maps use the numerical magnitudes in these units. Hessian-trace and differentiated-trace residuals consequently have units s^{-2} and $\mathrm{km}^{-1}\,\mathrm{s}^{-2}$, respectively. Statistics of different derivative orders are interpreted separately.

1 Tests No. 2–5: Supplementary polar-axis validation

This section supplements the polar-axis comparisons in the main article. Four source geometries are evaluated: Test No. 2, a normal tesseroid; Test No. 3, an extremely small tesseroid; Test No. 4, an extremely thin tesseroid; and Test No. 5, a complete zonal band. Their limits $(\varphi_1, \varphi_2, \lambda_1, \lambda_2, r_1, r_2)$ are, respectively,

$$
\begin{aligned}
\text{Test No. 2 (Normal):} &\quad (69.99^\circ, 89.99^\circ, -10^\circ, 10^\circ, 6341.2, 6371.2\ \mathrm{km}),\\
\text{Test No. 3 (Small):} &\quad (89.89^\circ, 89.99^\circ, -0.05^\circ, 0.05^\circ, 6370.2, 6371.2\ \mathrm{km}),\\
\text{Test No. 4 (Thin):} &\quad (69.99^\circ, 89.99^\circ, -10^\circ, 10^\circ, 6371.19, 6371.2\ \mathrm{km}),\\
\text{Test No. 5 (Zonal band):} &\quad (69.99^\circ, 89.99^\circ, -180^\circ, 180^\circ, 6341.2, 6371.2\ \mathrm{km}).
\end{aligned}
$$

The field is sampled at 2,001 points on the positive polar axis from $R_E - 1000$ km to $R_E + 1000$ km in 1-km increments, where $R_E = 6371.2$ km. The comparisons cover eight selected quantities: V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}. Finite-longitude sectors may also have other nonzero components at an observation on the polar axis. The analytical expressions are evaluated with 128-digit arithmetic, then rounded to IEEE 754 double precision for the archived error calculation; the computed responses also use double precision. The archived calculations use the normalized product $G\rho = 1$ s^{-2} in the kilometre-based unit system; this convention is consistent with the units stated above and is not an SI calculation with $G = 1$ and $\rho = 1$.

For separated observations at an exact geographic pole, the implementation used here provides a dedicated Cartesian path with analytical longitude integration and double-exponential quadrature of the remaining radial–polar integral. The archived figures compare the reported exact-axis responses with the analytical solution. They do not establish convergence of the general four-edge quadrature towards the dedicated polar path as the observation approaches a pole.

For each component X, the absolute and relative errors are

$$\epsilon_{\text{abs}} = |X_{\text{num}} - X_{\text{ana}}|, \qquad \epsilon_{\text{rel}} = \frac{|X_{\text{num}} - X_{\text{ana}}|}{\max(|X_{\text{ana}}|, \text{realmin}_{64})}. \tag{1}$$

Here realmin_{64} is the smallest positive normal binary64 value, the floor used in the archived calculation. Stored zero errors indicate equality of the rounded values. Figures S1–S4 show the complete relative-error profiles, Figs. S5–S8 give the corresponding absolute errors, and Figs. S9–S12 directly superpose the analytical and computed values. Absolute-error and direct-value profiles are needed when the reference approaches a zero crossing and the relative error is amplified.

1.1 Relative errors

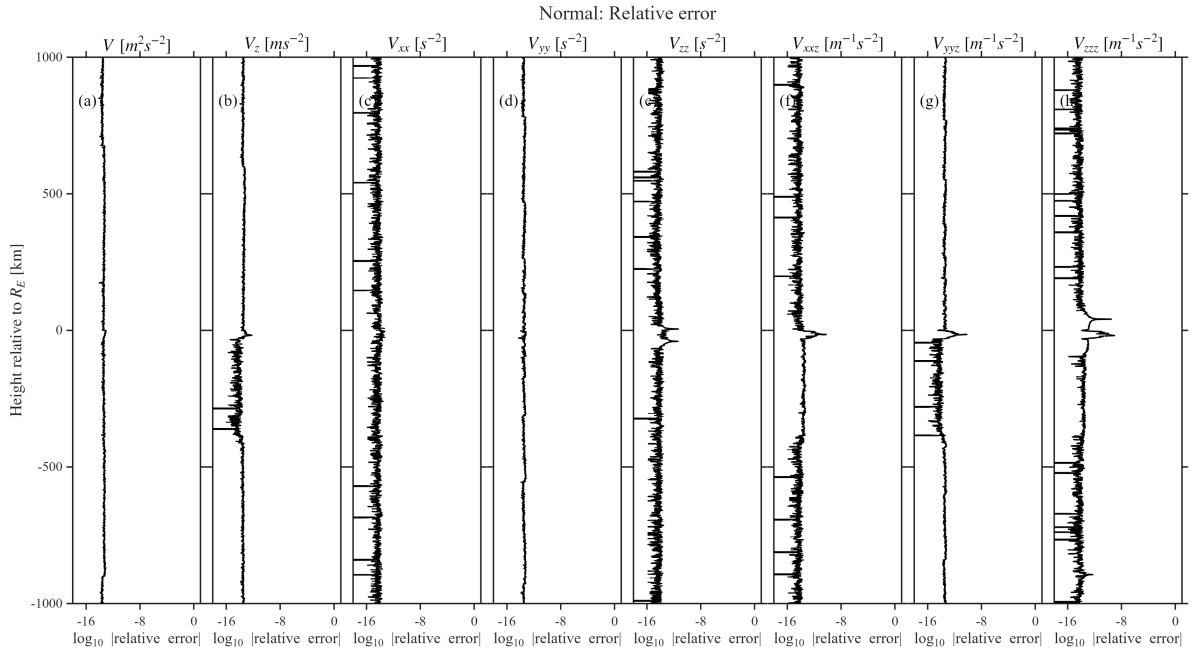


Figure S1: Test No. 2 (normal tesseroid): logarithmic relative errors of the computed polar-axis response against the analytical reference evaluated with 128 digits and rounded to double precision. Panels (a)–(h) show the eight selected quantities V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively.

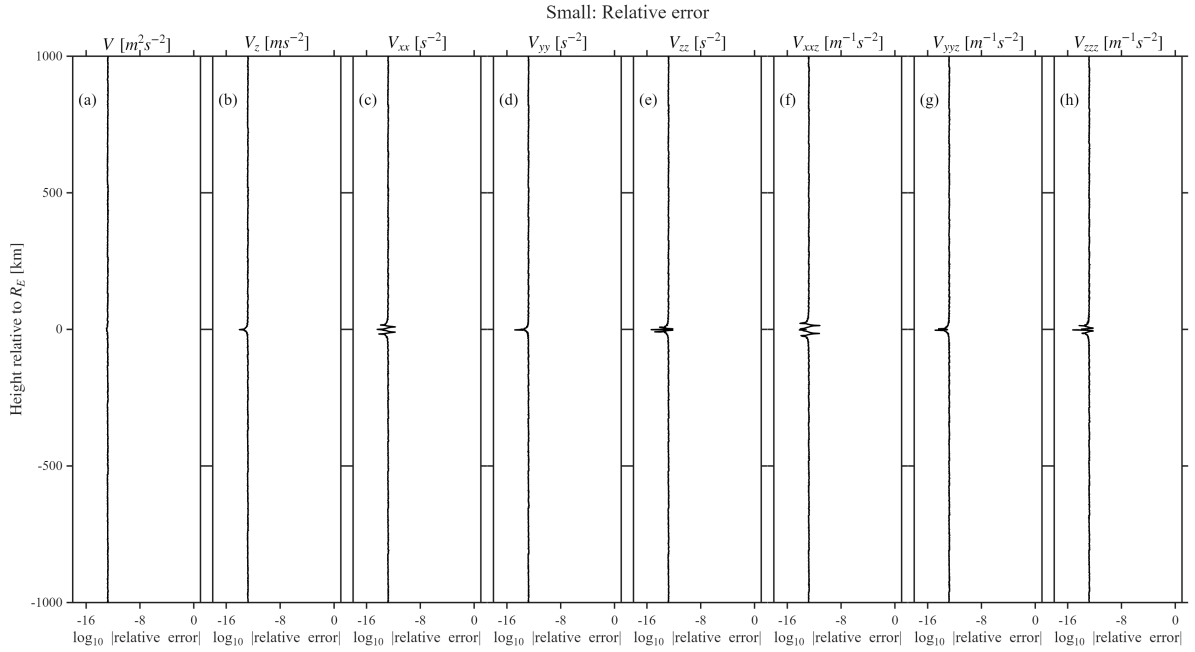


Figure S2: Test No. 3 (extremely small tesseroid): logarithmic relative errors along the positive polar axis. Panel order is the same as in Fig. S1.

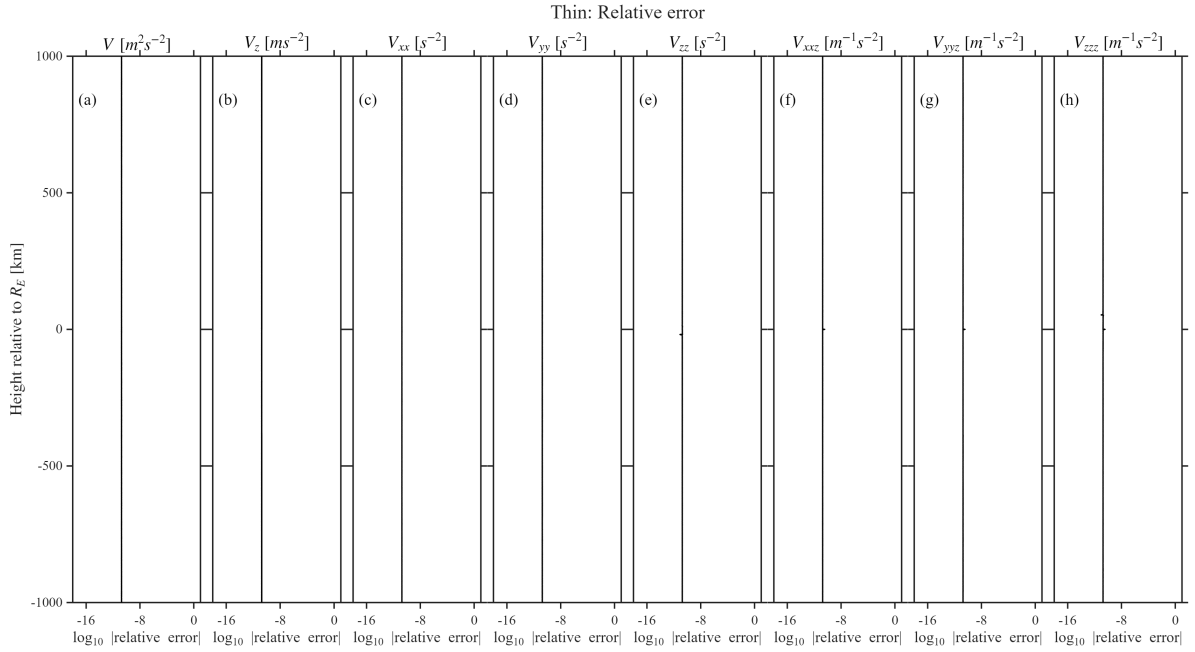


Figure S3: Test No. 4 (extremely thin tesseroid): logarithmic relative errors along the positive polar axis. The radial thickness is 10 m.

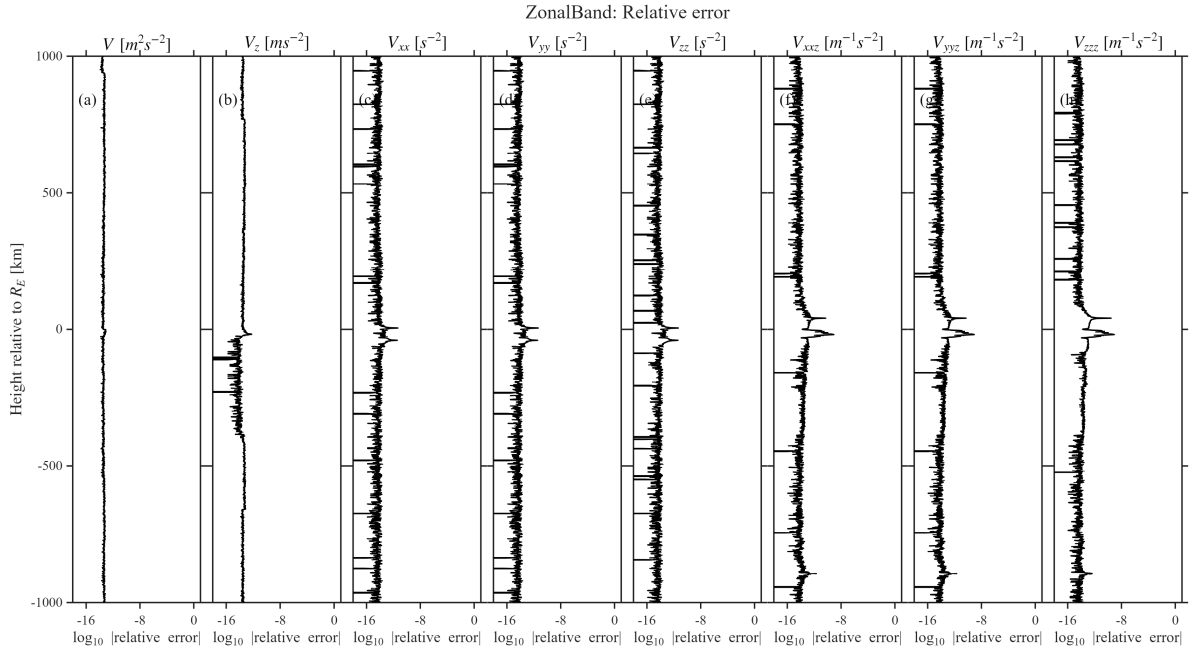


Figure S4: Test No. 5 (complete zonal band): logarithmic relative errors along the positive polar axis.

1.2 Absolute errors

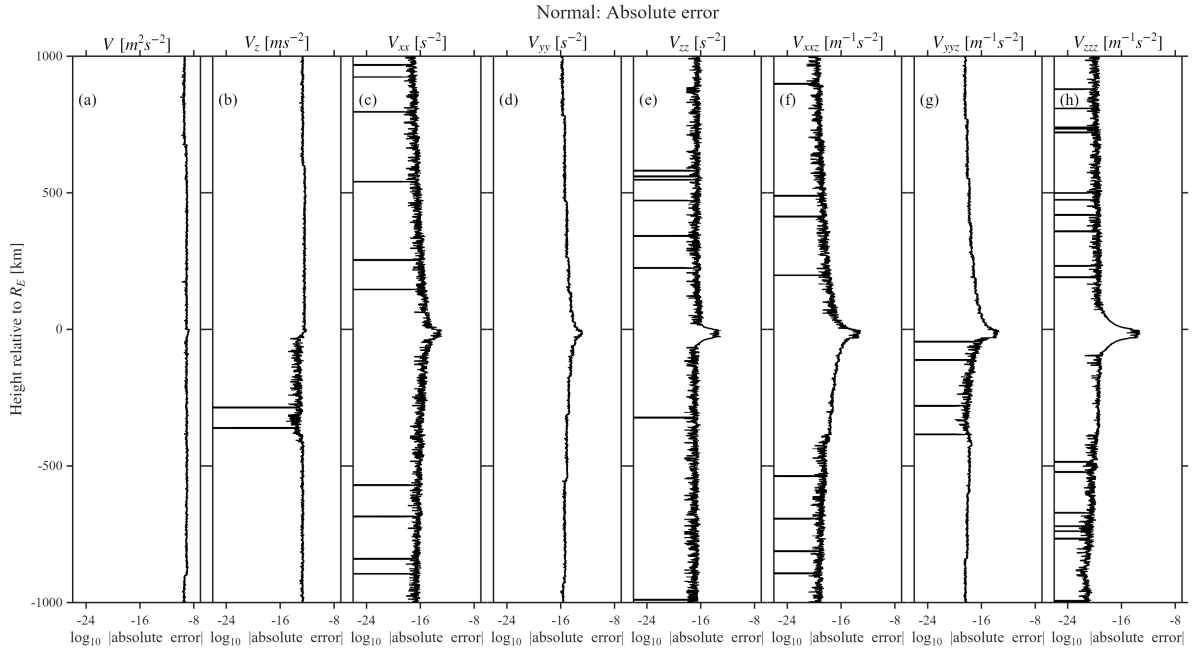


Figure S5: Test No. 2 (normal tesseroid): logarithmic absolute errors of the computed polar-axis response relative to the analytical reference. Panels (a)–(h) show V, V_z, V_{xx}, V_{yy}, V_{zz}, V_{xxz}, V_{yyz}, and V_{zzz}, respectively. The numerical error magnitudes use the kilometre-based units specified in the introduction.

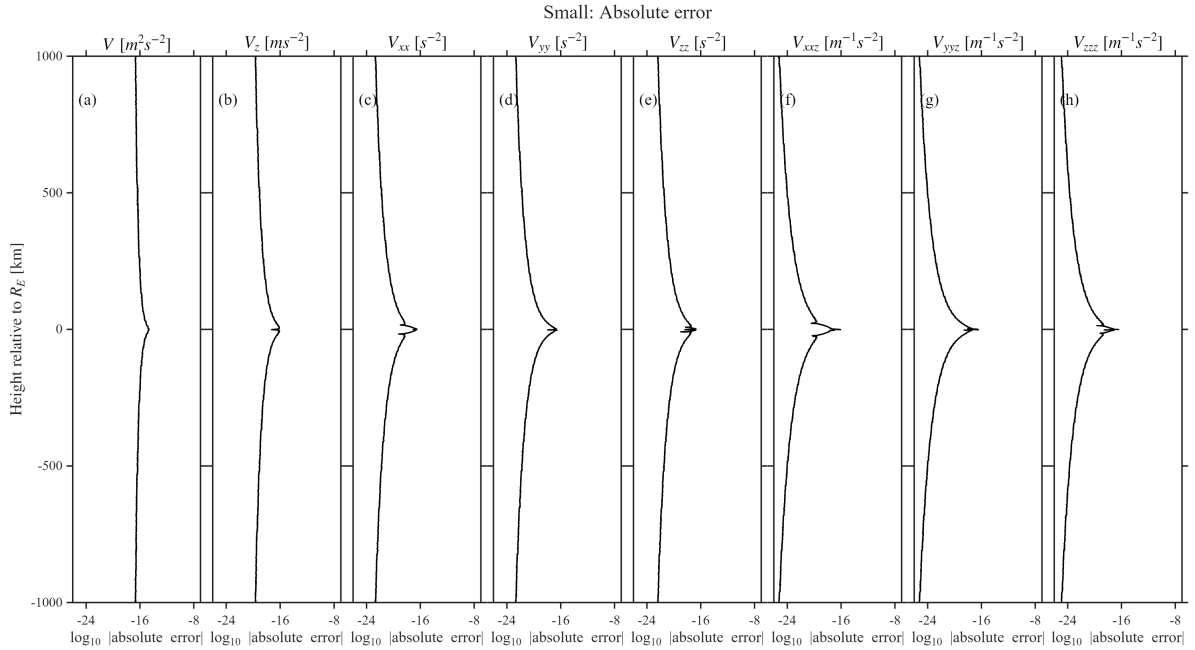


Figure S6: Test No. 3 (extremely small tesseroid): logarithmic absolute errors along the positive polar axis. Panel order is the same as in Fig. S5.

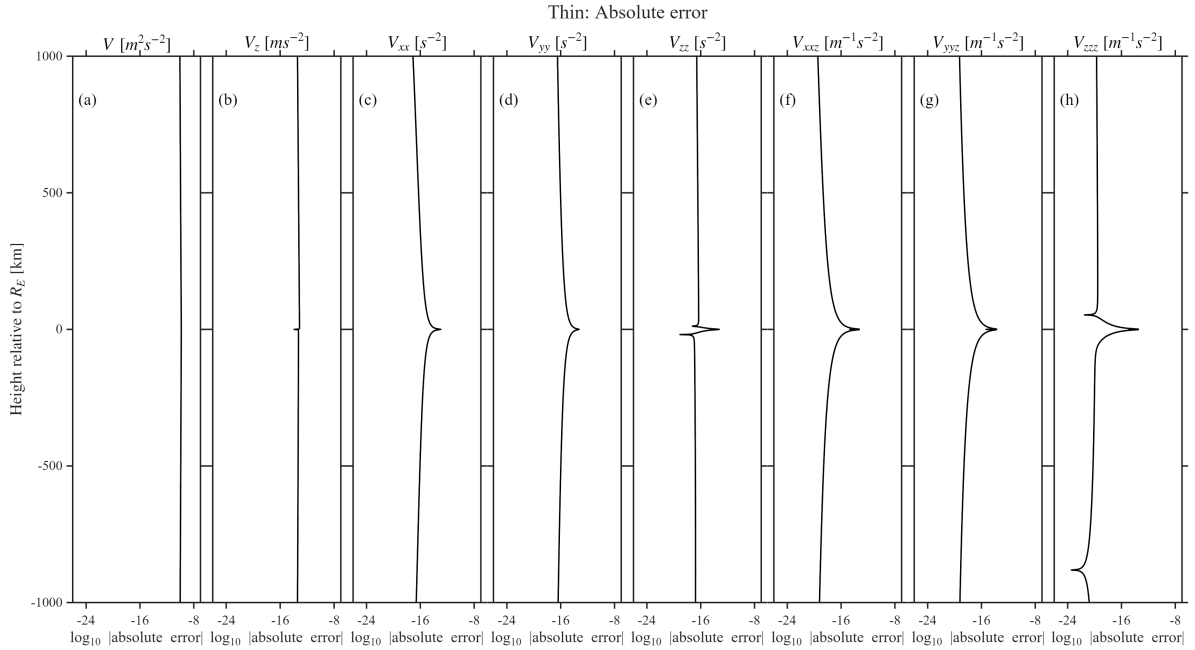


Figure S7: Test No. 4 (extremely thin tesseroid): logarithmic absolute errors along the positive polar axis. The radial thickness is 10 m.

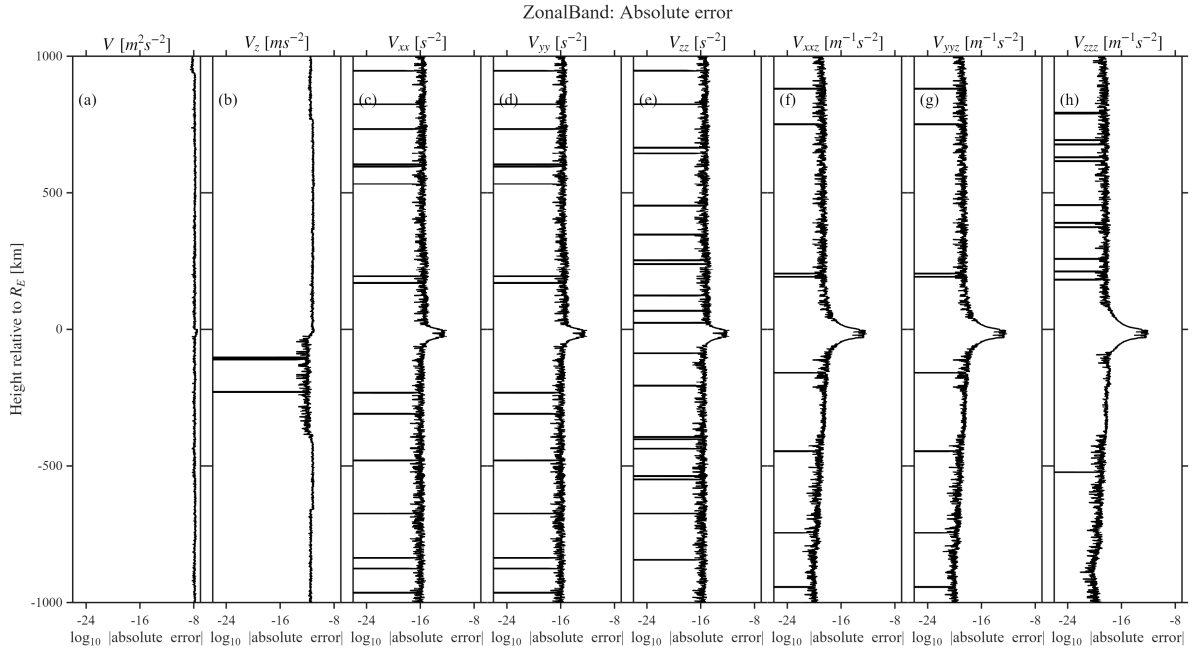


Figure S8: Test No. 5 (complete zonal band): logarithmic absolute errors along the positive polar axis.

1.3 Analytical and computed values

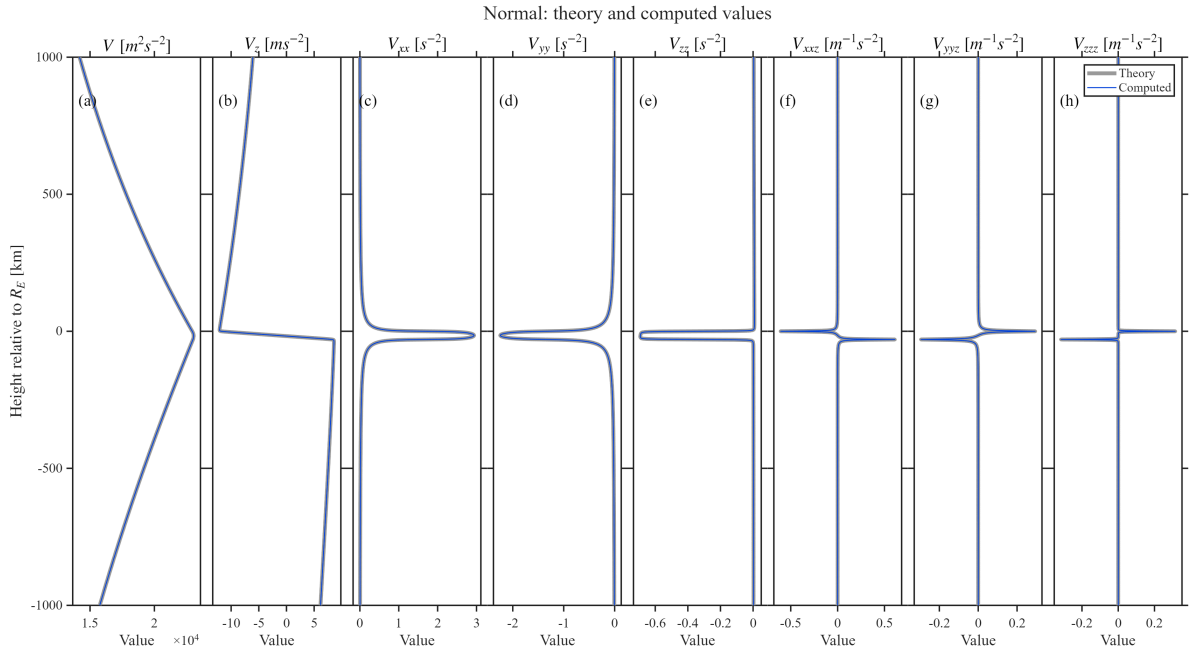


Figure S9: Test No. 2 (normal tesseroid): analytical values (grey) and computed values (blue) for the eight selected polar-axis quantities listed in Fig. S5. Both use the kilometre-based units specified in the introduction.

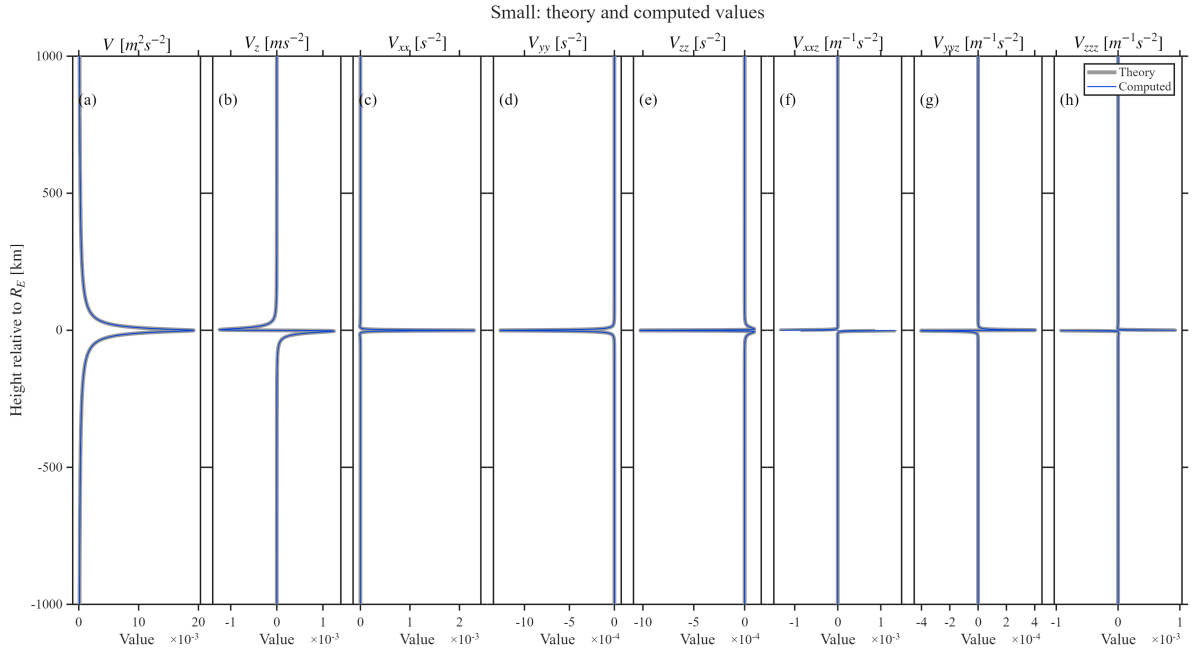


Figure S10: Test No. 3 (extremely small tesseroid): analytical values (grey) and computed polar-axis values (blue).

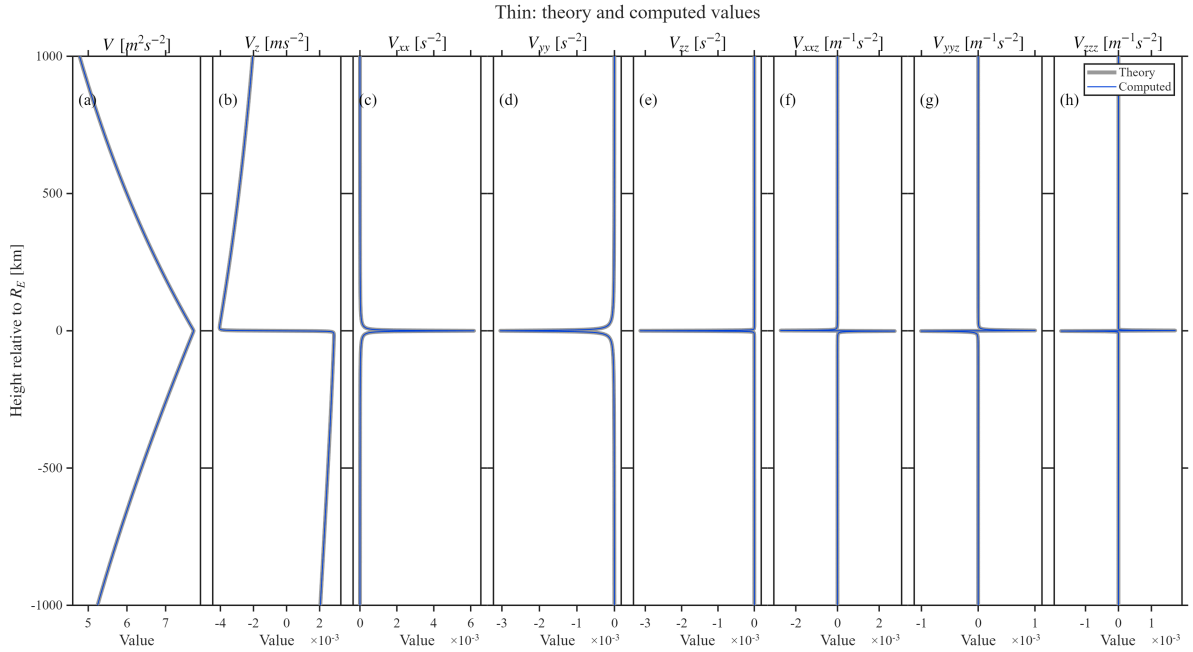


Figure S11: Test No. 4 (extremely thin tesseroid): analytical values (grey) and computed polar-axis values (blue).

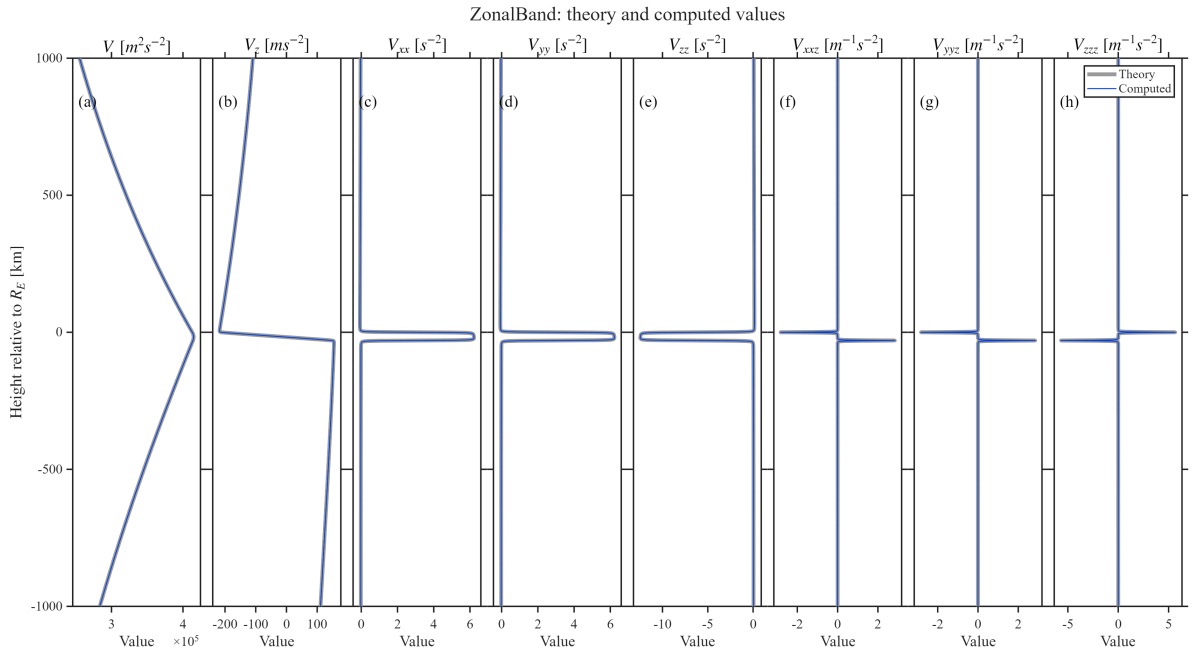


Figure S12: Test No. 5 (complete zonal band): analytical values (grey) and computed polar-axis values (blue).

2 Tests No. 6–8: Complete-spherical-shell validation

The complete-spherical-shell section of the main article defines the 30-km-thick shell, the global $1^\circ \times 1^\circ$ TLI–MMSI model and observation grid, and the three radial configurations, and presents Test No. 6 at $R_E + 1$ km. Figures S13 and S14 below show Test No. 7 at $R_E - 31$ km, in the cavity below the shell, and Test No. 8 at $R_E - 15$ km, within the shell material. These archived runs check spherical symmetry and the three radial regimes. Identical material states cancel at shared internal boundaries, so nonuniform interface jumps and near-/far-field matching require separate comparisons.

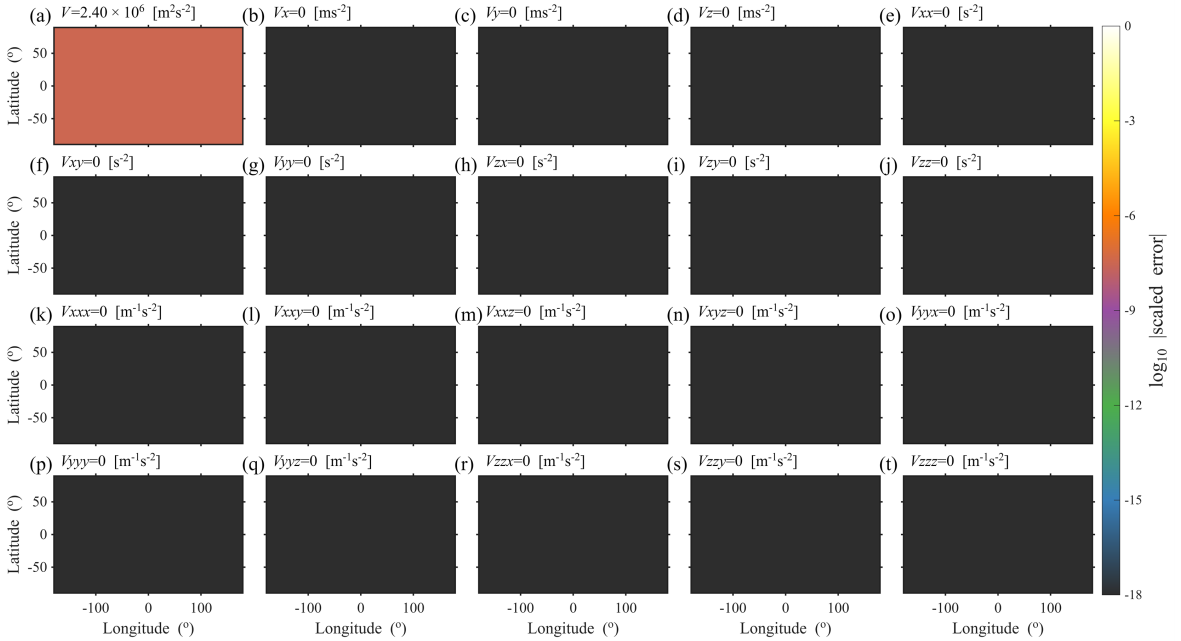


Figure S13: Test No. 7: complete-spherical-shell comparison at $R_E - 31$ km, 1 km below the inner shell boundary. The 20 panels map the base-10 logarithm of the absolute errors of the potential and its derivatives through third order over the global $1° \times 1°$ observation grid. Numerical error magnitudes use the kilometre-based units specified in the introduction.

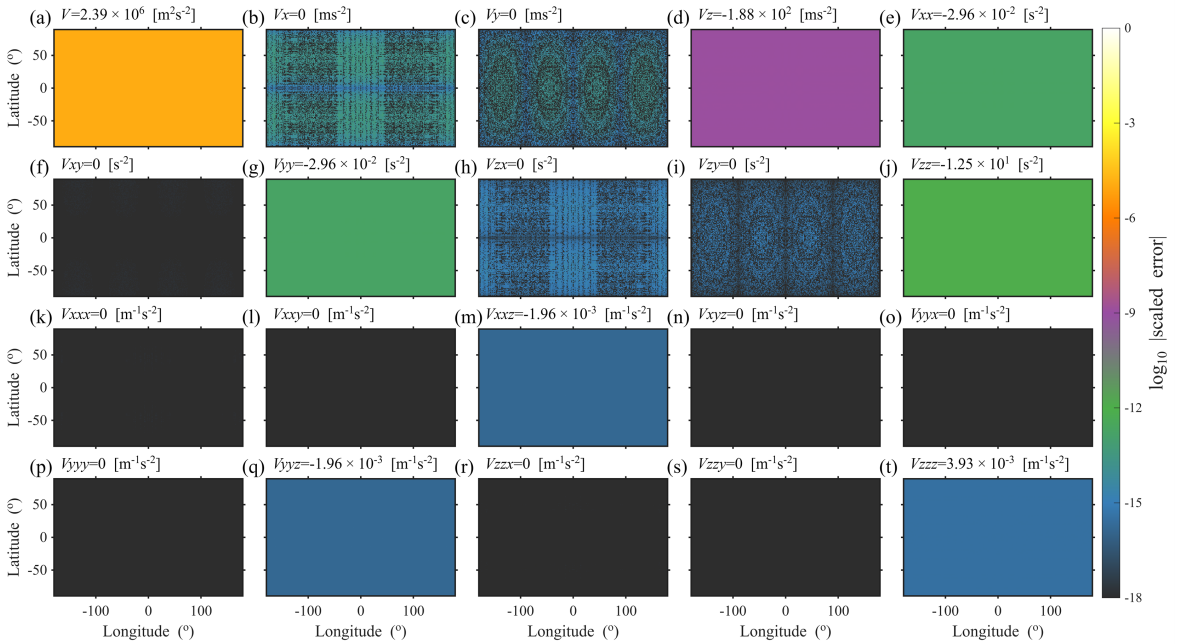


Figure S14: Test No. 8: complete-spherical-shell validation at $R_E - 15$ km, within the shell material. Panel order and error definition are the same as in Fig. S13; the analytical reference includes the interior Poisson contribution.

3 Tests No. 9–11: Laplace–Poisson and differentiated-trace validation

The Laplace–Poisson section of the main article defines the test tesseroid, the 401×401 observation grid, and the second- and third-order trace identities, and presents Test No. 9 at $R_E + 1$ km. Figures S15 and S16 give Test No. 10 at $R_E - 31$ km and Test No. 11 at $R_E - 15$ km. The offset observation grids exclude points on material boundaries. These four identities assess differential consistency, complementing the componentwise reference comparisons.

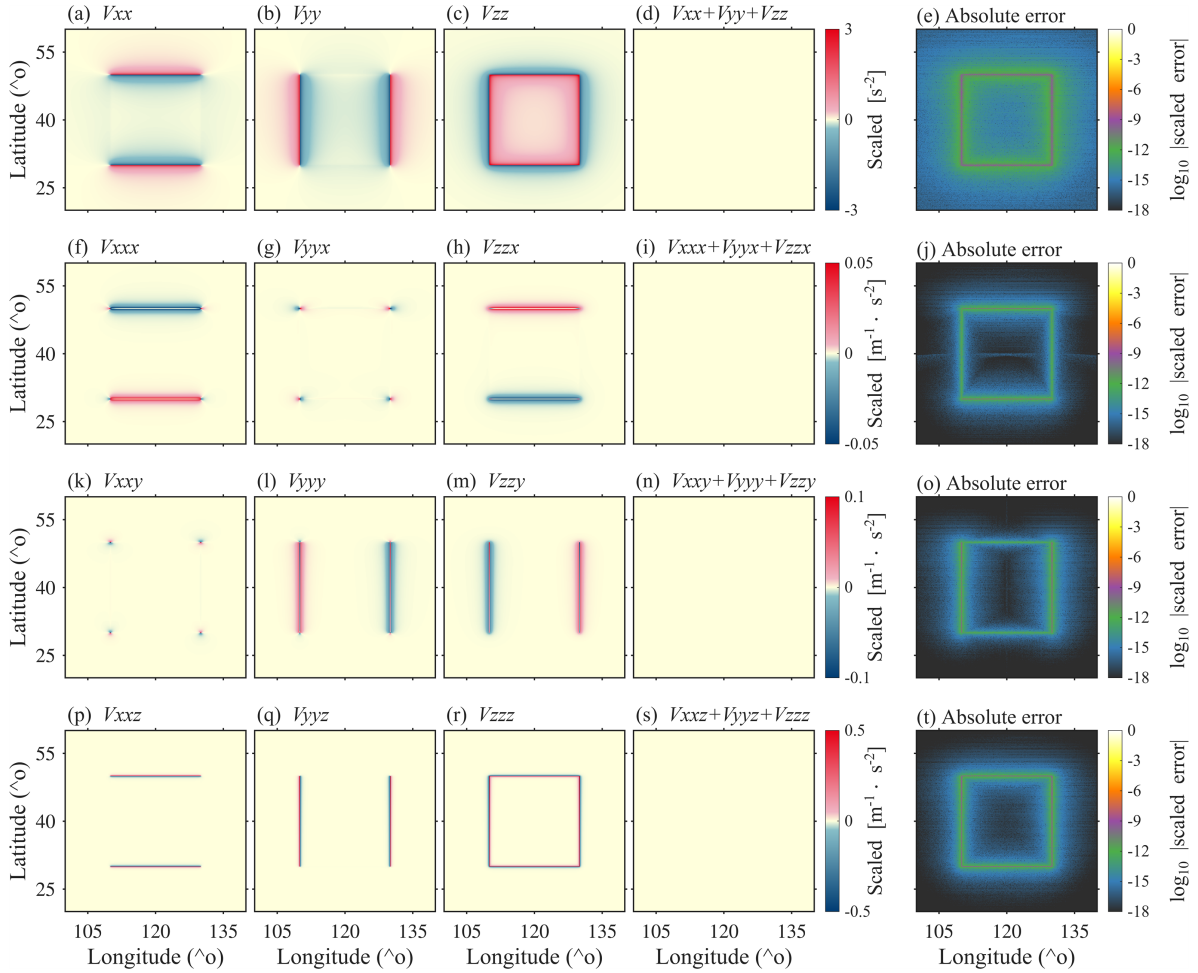


Figure S15: Test No. 10: Laplace and differentiated-trace checks at $R_E - 31$ km, 1 km below the test tesseroid. The four rows show the Hessian trace and its x-, y-, and z-derivative identities. Columns 1–3 show the individual tensor components, column 4 their sum, and column 5 the base-10 logarithm of the absolute residual. Numerical Hessian-trace and differentiated-trace residuals use units s^{-2} and $\mathrm{km}^{-1}\,\mathrm{s}^{-2}$, respectively.

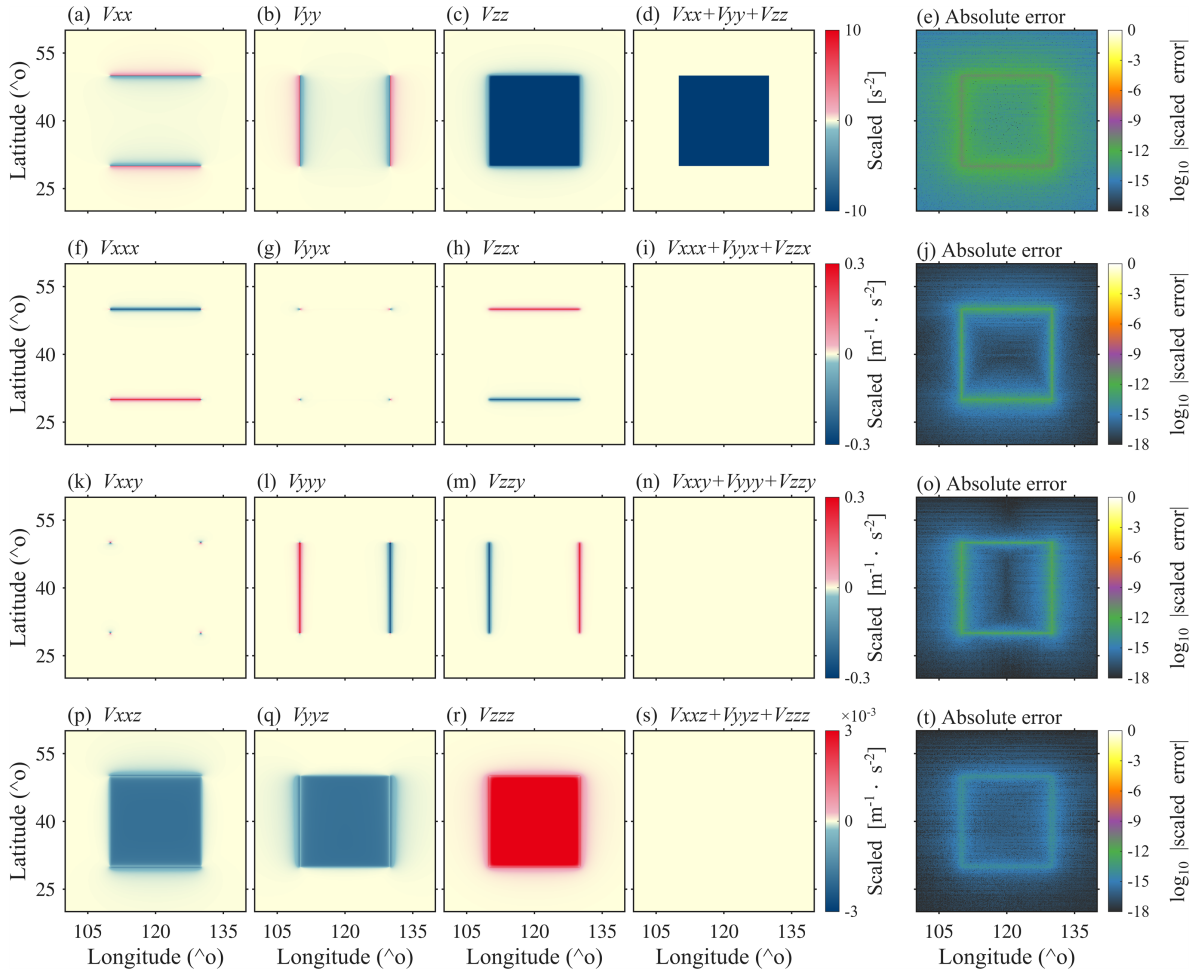


Figure S16: Test No. 11: Laplace–Poisson and differentiated-trace validation at $R_E - 15$ km, within the radial extent of the test tesseroid. Panel order is the same as in Fig. S15; the theoretical Hessian trace is $-4\pi G\rho$ inside the angular footprint and zero outside it.

4 Statistical audit of Tests No. 1–11

The following tables are not manually transcribed. They are generated by the statistical audit script from the same archived result files used for Figs. S1–S16 and for the numerical statements in the main article. The human-readable audit log prints every manuscript-level extremum and timing claim, while machine-readable CSV files preserve both componentwise and pooled experiment statistics.

4.1 Reproducible statistical audit of Tests No. 1–11

All entries below are recomputed from the archived pointwise outputs. For a nonnegative error sample e_i, RMSE is $[N^{-1}\sum_i e_i^2]^{1/2}$; P50, P90, and P99 are empirical percentiles. No finite value is removed. The accompanying statistical audit reports the count of nonfinite values separately.

Table S1: Statistics of the componentwise relative differences in Test No. 1.

Height	Comparison	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
100 km	2-D vs 3-D	20	2.36×10^{-28}	5.55×10^{-24}	4.63×10^{-25}	1.45×10^{-24}	5.24×10^{-27}	1.77×10^{-24}	5.55×10^{-24}
100 km	1-D vs 3-D	20	3.21×10^{-26}	1.12×10^{-22}	1.29×10^{-23}	3.25×10^{-23}	4.57×10^{-25}	6.20×10^{-23}	1.12×10^{-22}
100 km	double-precision TLI vs 3-D	20	4.13×10^{-14}	7.76×10^{-13}	2.55×10^{-13}	3.20×10^{-13}	1.98×10^{-13}	5.94×10^{-13}	7.76×10^{-13}
400 km	2-D vs 3-D	20	6.39×10^{-28}	1.05×10^{-23}	1.62×10^{-24}	3.44×10^{-24}	1.43×10^{-26}	6.77×10^{-24}	1.05×10^{-23}
400 km	1-D vs 3-D	20	3.48×10^{-27}	1.26×10^{-21}	1.24×10^{-22}	3.21×10^{-22}	7.46×10^{-25}	4.60×10^{-22}	1.26×10^{-21}
400 km	double-precision TLI vs 3-D	20	1.50×10^{-14}	4.16×10^{-13}	9.90×10^{-14}	1.28×10^{-13}	8.89×10^{-14}	1.52×10^{-13}	4.16×10^{-13}

Table S2: Componentwise absolute- and relative-error statistics for Test No. 2 over 2,001 polar-axis points.

Component	Error	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	absolute	2001	1.22×10^{-10}	2.02×10^{-9}	8.00×10^{-10}	8.44×10^{-10}	8.44×10^{-10}	1.09×10^{-9}	1.39×10^{-9}
V	relative	2001	8.05×10^{-15}	8.78×10^{-14}	4.24×10^{-14}	4.43×10^{-14}	4.33×10^{-14}	5.81×10^{-14}	7.12×10^{-14}
V_z	absolute	2001	0	7.87×10^{-13}	2.54×10^{-13}	2.91×10^{-13}	2.30×10^{-13}	4.69×10^{-13}	5.96×10^{-13}
V_z	relative	2001	0	6.37×10^{-13}	3.18×10^{-14}	3.94×10^{-14}	3.20×10^{-14}	4.94×10^{-14}	7.04×10^{-14}
V_{xx}	absolute	2001	0	1.40×10^{-13}	1.12×10^{-15}	8.23×10^{-15}	5.55×10^{-17}	5.83×10^{-16}	3.95×10^{-14}
V_{xx}	relative	2001	0	5.26×10^{-14}	6.42×10^{-15}	8.34×10^{-15}	5.28×10^{-15}	1.32×10^{-14}	2.25×10^{-14}
V_{yy}	absolute	2001	7.81×10^{-17}	1.63×10^{-13}	2.96×10^{-15}	1.32×10^{-14}	5.95×10^{-16}	3.71×10^{-15}	6.87×10^{-14}
V_{yy}	relative	2001	5.57×10^{-15}	7.42×10^{-14}	3.78×10^{-14}	3.95×10^{-14}	3.63×10^{-14}	5.37×10^{-14}	6.73×10^{-14}
V_{zz}	absolute	2001	0	8.75×10^{-14}	4.36×10^{-16}	4.04×10^{-15}	2.26×10^{-17}	6.94×10^{-17}	1.53×10^{-14}
V_{zz}	relative	2001	0	4.63×10^{-12}	1.42×10^{-14}	1.42×10^{-13}	5.62×10^{-15}	1.57×10^{-14}	1.19×10^{-13}
V_{xxz}	absolute	2001	0	7.06×10^{-14}	4.74×10^{-16}	4.37×10^{-15}	2.17×10^{-19}	1.70×10^{-17}	1.75×10^{-14}
V_{xxz}	relative	2001	0	6.09×10^{-11}	7.39×10^{-14}	1.50×10^{-12}	6.88×10^{-15}	2.95×10^{-14}	3.74×10^{-13}
V_{yyz}	absolute	2001	0	4.01×10^{-14}	2.45×10^{-16}	2.04×10^{-15}	1.07×10^{-18}	1.70×10^{-17}	9.54×10^{-15}
V_{yyz}	relative	2001	0	6.82×10^{-11}	7.92×10^{-14}	1.54×10^{-12}	3.38×10^{-14}	5.08×10^{-14}	2.16×10^{-13}
V_{zzz}	absolute	2001	0	5.49×10^{-14}	3.09×10^{-16}	3.02×10^{-15}	1.86×10^{-20}	8.95×10^{-20}	1.06×10^{-14}
V_{zzz}	relative	2001	0	8.88×10^{-10}	1.39×10^{-12}	2.48×10^{-11}	8.03×10^{-15}	3.55×10^{-14}	4.93×10^{-12}

Table S3: Componentwise absolute- and relative-error statistics for Test No. 3 over 2,001 polar-axis points.

Component	Error	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	absolute	2001	1.92×10^{-17}	2.26×10^{-15}	1.09×10^{-16}	2.52×10^{-16}	4.40×10^{-17}	2.13×10^{-16}	1.47×10^{-15}
V	relative	2001	1.02×10^{-13}	2.01×10^{-13}	1.62×10^{-13}	1.63×10^{-13}	1.62×10^{-13}	1.78×10^{-13}	1.89×10^{-13}
V_z	absolute	2001	1.84×10^{-20}	8.22×10^{-17}	1.98×10^{-18}	8.78×10^{-18}	8.78×10^{-20}	1.92×10^{-18}	5.60×10^{-17}
V_z	relative	2001	9.08×10^{-15}	2.00×10^{-13}	1.60×10^{-13}	1.61×10^{-13}	1.60×10^{-13}	1.77×10^{-13}	1.89×10^{-13}
V_{xx}	absolute	2001	1.89×10^{-23}	3.17×10^{-17}	1.54×10^{-19}	1.56×10^{-18}	1.72×10^{-22}	1.96×10^{-20}	3.41×10^{-18}
V_{xx}	relative	2001	3.78×10^{-15}	1.91×10^{-12}	1.62×10^{-13}	1.74×10^{-13}	1.60×10^{-13}	1.76×10^{-13}	1.90×10^{-13}
V_{yy}	absolute	2001	1.88×10^{-23}	2.98×10^{-17}	1.77×10^{-19}	1.40×10^{-18}	1.74×10^{-22}	2.10×10^{-20}	5.59×10^{-18}
V_{yy}	relative	2001	1.48×10^{-15}	1.98×10^{-13}	1.60×10^{-13}	1.61×10^{-13}	1.60×10^{-13}	1.76×10^{-13}	1.89×10^{-13}
V_{zz}	absolute	2001	3.88×10^{-23}	1.99×10^{-17}	1.37×10^{-19}	1.05×10^{-18}	3.54×10^{-22}	4.14×10^{-20}	2.82×10^{-18}
V_{zz}	relative	2001	4.23×10^{-16}	7.16×10^{-13}	1.60×10^{-13}	1.62×10^{-13}	1.60×10^{-13}	1.77×10^{-13}	1.90×10^{-13}
V_{xxz}	absolute	2001	5.45×10^{-26}	8.39×10^{-17}	9.45×10^{-20}	2.13×10^{-18}	9.46×10^{-25}	5.34×10^{-22}	1.04×10^{-18}
V_{xxz}	relative	2001	6.15×10^{-15}	7.08×10^{-12}	1.66×10^{-13}	2.65×10^{-13}	1.60×10^{-13}	1.76×10^{-13}	1.92×10^{-13}
V_{yyz}	absolute	2001	5.59×10^{-26}	3.52×10^{-17}	7.17×10^{-20}	1.20×10^{-18}	9.52×10^{-25}	6.69×10^{-22}	7.53×10^{-19}
V_{yyz}	relative	2001	1.26×10^{-15}	2.01×10^{-13}	1.59×10^{-13}	1.60×10^{-13}	1.60×10^{-13}	1.76×10^{-13}	1.89×10^{-13}
V_{zzz}	absolute	2001	1.15×10^{-25}	3.98×10^{-17}	5.93×10^{-20}	1.08×10^{-18}	1.86×10^{-24}	1.14×10^{-21}	4.91×10^{-19}
V_{zzz}	relative	2001	5.88×10^{-16}	6.20×10^{-13}	1.59×10^{-13}	1.62×10^{-13}	1.59×10^{-13}	1.77×10^{-13}	1.90×10^{-13}

Table S4: Componentwise absolute- and relative-error statistics for Test No. 4 over 2,001 polar-axis points.

Component	Error	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	absolute	2001	8.71×10^{-11}	1.41×10^{-10}	1.14×10^{-10}	1.15×10^{-10}	1.13×10^{-10}	1.35×10^{-10}	1.40×10^{-10}
V	relative	2001	1.82×10^{-11}	1.83×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.83×10^{-11}
V_z	absolute	2001	1.16×10^{-14}	7.39×10^{-14}	4.95×10^{-14}	5.04×10^{-14}	4.77×10^{-14}	6.49×10^{-14}	7.33×10^{-14}
V_z	relative	2001	1.81×10^{-11}	1.87×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.83×10^{-11}
V_{xx}	absolute	2001	7.96×10^{-18}	1.12×10^{-13}	4.77×10^{-16}	3.63×10^{-15}	7.23×10^{-17}	5.65×10^{-16}	6.20×10^{-15}
V_{xx}	relative	2001	1.81×10^{-11}	1.83×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}
V_{yy}	absolute	2001	3.33×10^{-17}	5.52×10^{-14}	4.40×10^{-16}	2.32×10^{-15}	9.80×10^{-17}	5.95×10^{-16}	6.17×10^{-15}
V_{yy}	relative	2001	1.81×10^{-11}	1.83×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.83×10^{-11}
V_{zz}	absolute	2001	8.20×10^{-20}	5.69×10^{-14}	8.90×10^{-17}	1.49×10^{-15}	2.54×10^{-17}	4.51×10^{-17}	5.18×10^{-17}
V_{zz}	relative	2001	8.55×10^{-12}	1.92×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.84×10^{-11}
V_{xxz}	absolute	2001	3.44×10^{-20}	5.00×10^{-14}	9.44×10^{-17}	1.70×10^{-15}	2.36×10^{-19}	6.20×10^{-18}	6.33×10^{-16}
V_{xxz}	relative	2001	1.81×10^{-11}	3.52×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}
V_{yyz}	absolute	2001	5.43×10^{-20}	1.82×10^{-14}	5.12×10^{-17}	7.03×10^{-16}	2.46×10^{-19}	6.23×10^{-18}	5.58×10^{-16}
V_{yyz}	relative	2001	1.81×10^{-11}	3.52×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.83×10^{-11}
V_{zzz}	absolute	2001	3.44×10^{-24}	3.18×10^{-14}	4.33×10^{-17}	1.03×10^{-15}	2.04×10^{-20}	2.91×10^{-20}	2.30×10^{-17}
V_{zzz}	relative	2001	9.89×10^{-12}	3.52×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.82×10^{-11}	1.84×10^{-11}

Table S5: Componentwise absolute- and relative-error statistics for Test No. 5 over 2,001 polar-axis points.

Component	Error	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	absolute	2001	3.23×10^{-9}	3.63×10^{-8}	1.41×10^{-8}	1.45×10^{-8}	1.41×10^{-8}	1.79×10^{-8}	2.25×10^{-8}
V	relative	2001	1.24×10^{-14}	8.75×10^{-14}	4.25×10^{-14}	4.39×10^{-14}	4.22×10^{-14}	5.65×10^{-14}	6.85×10^{-14}
V_z	absolute	2001	0	1.23×10^{-11}	4.91×10^{-12}	5.57×10^{-12}	4.82×10^{-12}	8.30×10^{-12}	1.02×10^{-11}
V_z	relative	2001	0	6.10×10^{-13}	3.45×10^{-14}	4.24×10^{-14}	3.31×10^{-14}	5.65×10^{-14}	7.21×10^{-14}
V_{xx}	absolute	2001	0	7.27×10^{-13}	3.72×10^{-15}	3.41×10^{-14}	1.87×10^{-16}	6.66×10^{-16}	1.26×10^{-13}
V_{xx}	relative	2001	0	4.52×10^{-12}	1.36×10^{-14}	1.34×10^{-13}	5.60×10^{-15}	1.51×10^{-14}	1.16×10^{-13}
V_{yy}	absolute	2001	0	7.27×10^{-13}	3.72×10^{-15}	3.41×10^{-14}	1.87×10^{-16}	6.66×10^{-16}	1.26×10^{-13}
V_{yy}	relative	2001	0	4.52×10^{-12}	1.36×10^{-14}	1.34×10^{-13}	5.60×10^{-15}	1.51×10^{-14}	1.16×10^{-13}
V_{zz}	absolute	2001	0	1.45×10^{-12}	7.42×10^{-15}	6.81×10^{-14}	3.68×10^{-16}	1.31×10^{-15}	2.56×10^{-13}
V_{zz}	relative	2001	0	4.60×10^{-12}	1.39×10^{-14}	1.43×10^{-13}	5.67×10^{-15}	1.48×10^{-14}	1.16×10^{-13}
V_{xxz}	absolute	2001	0	5.27×10^{-13}	2.86×10^{-15}	2.76×10^{-14}	1.64×10^{-19}	1.07×10^{-18}	1.12×10^{-13}
V_{xxz}	relative	2001	0	8.50×10^{-10}	1.32×10^{-12}	2.46×10^{-11}	9.06×10^{-15}	4.68×10^{-14}	4.23×10^{-12}
V_{yyz}	absolute	2001	0	5.27×10^{-13}	2.86×10^{-15}	2.76×10^{-14}	1.64×10^{-19}	1.07×10^{-18}	1.12×10^{-13}
V_{yyz}	relative	2001	0	8.50×10^{-10}	1.32×10^{-12}	2.46×10^{-11}	9.06×10^{-15}	4.68×10^{-14}	4.23×10^{-12}
V_{zzz}	absolute	2001	0	1.05×10^{-12}	5.73×10^{-15}	5.51×10^{-14}	3.35×10^{-19}	1.63×10^{-18}	2.28×10^{-13}
V_{zzz}	relative	2001	0	8.52×10^{-10}	1.44×10^{-12}	2.55×10^{-11}	7.82×10^{-15}	3.89×10^{-14}	4.26×10^{-12}

Table S6: Componentwise absolute-error statistics for the complete-spherical-shell Test No. 6 over 64,800 points.

Component	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	64800	3.50×10^{-7}	3.52×10^{-7}	3.51×10^{-7}	3.51×10^{-7}	3.51×10^{-7}	3.51×10^{-7}	3.52×10^{-7}
V_x	64800	0	1.34×10^{-13}	1.82×10^{-14}	2.75×10^{-14}	1.34×10^{-14}	5.35×10^{-14}	8.03×10^{-14}
V_y	64800	0	5.35×10^{-14}	5.75×10^{-15}	1.09×10^{-14}	0	2.68×10^{-14}	2.68×10^{-14}
V_z	64800	5.47×10^{-11}	5.53×10^{-11}	5.50×10^{-11}	5.50×10^{-11}	5.50×10^{-11}	5.51×10^{-11}	5.51×10^{-11}
V_{xx}	64800	8.55×10^{-15}	8.71×10^{-15}	8.63×10^{-15}	8.63×10^{-15}	8.63×10^{-15}	8.65×10^{-15}	8.69×10^{-15}
V_{xy}	64800	0	6.54×10^{-18}	6.24×10^{-19}	1.29×10^{-18}	1.12×10^{-34}	3.27×10^{-18}	3.27×10^{-18}
V_{xz}	64800	0	4.67×10^{-17}	7.00×10^{-18}	10.00×10^{-18}	4.67×10^{-18}	1.59×10^{-17}	3.13×10^{-17}
V_{yy}	64800	8.56×10^{-15}	8.71×10^{-15}	8.63×10^{-15}	8.63×10^{-15}	8.63×10^{-15}	8.66×10^{-15}	8.69×10^{-15}
V_{yz}	64800	0	2.52×10^{-17}	2.72×10^{-18}	4.73×10^{-18}	8.17×10^{-19}	9.33×10^{-18}	1.26×10^{-17}
V_{zz}	64800	1.71×10^{-14}	1.74×10^{-14}	1.73×10^{-14}	1.73×10^{-14}	1.73×10^{-14}	1.73×10^{-14}	1.74×10^{-14}
V_{xxx}	64800	0	2.97×10^{-20}	4.04×10^{-21}	6.09×10^{-21}	2.97×10^{-21}	1.19×10^{-20}	1.78×10^{-20}
V_{xxy}	64800	0	3.96×10^{-21}	4.25×10^{-22}	8.03×10^{-22}	1.10×10^{-37}	1.98×10^{-21}	1.98×10^{-21}
V_{xxz}	64800	4.02×10^{-18}	4.10×10^{-18}	4.06×10^{-18}	4.06×10^{-18}	4.06×10^{-18}	4.08×10^{-18}	4.09×10^{-18}
V_{xyy}	64800	0	9.89×10^{-21}	1.35×10^{-21}	2.03×10^{-21}	9.89×10^{-22}	3.96×10^{-21}	5.93×10^{-21}
V_{xyz}	64800	0	3.08×10^{-21}	2.94×10^{-22}	6.06×10^{-22}	1.10×10^{-37}	1.54×10^{-21}	1.54×10^{-21}
V_{xzz}	64800	0	2.55×10^{-20}	3.93×10^{-21}	5.56×10^{-21}	3.08×10^{-21}	9.67×10^{-21}	1.76×10^{-20}
V_{yyy}	64800	0	1.19×10^{-20}	1.27×10^{-21}	2.41×10^{-21}	0	5.93×10^{-21}	5.93×10^{-21}
V_{yyz}	64800	4.02×10^{-18}	4.10×10^{-18}	4.06×10^{-18}	4.06×10^{-18}	4.06×10^{-18}	4.08×10^{-18}	4.09×10^{-18}
V_{yzz}	64800	0	1.58×10^{-20}	1.71×10^{-21}	2.93×10^{-21}	4.94×10^{-22}	4.83×10^{-21}	7.91×10^{-21}
V_{zzz}	64800	7.98×10^{-18}	8.28×10^{-18}	8.12×10^{-18}	8.12×10^{-18}	8.12×10^{-18}	8.16×10^{-18}	8.21×10^{-18}

Table S7: Componentwise absolute-error statistics for the complete-spherical-shell Test No. 7 over 64,800 points.

Component	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	64800	3.45×10^{-8}	3.45×10^{-8}	3.45×10^{-8}	3.45×10^{-8}	3.45×10^{-8}	3.45×10^{-8}	3.45×10^{-8}
V_x	64800	0	0	0	0	0	0	0
V_y	64800	0	0	0	0	0	0	0
V_z	64800	0	0	0	0	0	0	0
V_{xx}	64800	0	0	0	0	0	0	0
V_{xy}	64800	0	0	0	0	0	0	0
V_{xz}	64800	0	0	0	0	0	0	0
V_{yy}	64800	0	0	0	0	0	0	0
V_{yz}	64800	0	0	0	0	0	0	0
V_{zz}	64800	0	0	0	0	0	0	0
V_{xxx}	64800	0	0	0	0	0	0	0
V_{xxy}	64800	0	0	0	0	0	0	0
V_{xxz}	64800	0	0	0	0	0	0	0
V_{xyy}	64800	0	0	0	0	0	0	0
V_{xyz}	64800	0	0	0	0	0	0	0
V_{xzz}	64800	0	0	0	0	0	0	0
V_{yyy}	64800	0	0	0	0	0	0	0
V_{yyz}	64800	0	0	0	0	0	0	0
V_{yzz}	64800	0	0	0	0	0	0	0
V_{zzz}	64800	0	0	0	0	0	0	0

Table S8: Componentwise absolute-error statistics for the complete-spherical-shell Test No. 8 over 64,800 points.

Component	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
V	64800	1.17×10^{-5}	1.17×10^{-5}	1.17×10^{-5}	1.17×10^{-5}	1.17×10^{-5}	1.17×10^{-5}	1.17×10^{-5}
V_x	64800	0	8.07×10^{-14}	9.16×10^{-15}	1.39×10^{-14}	6.73×10^{-15}	2.69×10^{-14}	4.04×10^{-14}
V_y	64800	0	2.69×10^{-14}	2.87×10^{-15}	5.43×10^{-15}	0	1.35×10^{-14}	1.35×10^{-14}
V_z	64800	1.21×10^{-9}	1.30×10^{-9}	1.25×10^{-9}	1.25×10^{-9}	1.25×10^{-9}	1.26×10^{-9}	1.28×10^{-9}
V_{xx}	64800	1.90×10^{-13}	2.03×10^{-13}	1.96×10^{-13}	1.96×10^{-13}	1.96×10^{-13}	1.99×10^{-13}	2.00×10^{-13}
V_{xy}	64800	0	3.28×10^{-18}	3.14×10^{-19}	6.47×10^{-19}	7.98×10^{-33}	1.64×10^{-18}	1.64×10^{-18}
V_{xz}	64800	0	5.37×10^{-15}	6.09×10^{-16}	9.24×10^{-16}	4.47×10^{-16}	1.79×10^{-15}	2.68×10^{-15}
V_{yy}	64800	1.90×10^{-13}	2.04×10^{-13}	1.96×10^{-13}	1.96×10^{-13}	1.96×10^{-13}	1.99×10^{-13}	2.00×10^{-13}
V_{yz}	64800	0	1.79×10^{-15}	1.91×10^{-16}	3.61×10^{-16}	1.64×10^{-18}	8.93×10^{-16}	8.94×10^{-16}
V_{zz}	64800	1.19×10^{-12}	1.22×10^{-12}	1.21×10^{-12}	1.21×10^{-12}	1.21×10^{-12}	1.21×10^{-12}	1.21×10^{-12}
V_{xxx}	64800	0	2.53×10^{-18}	2.87×10^{-19}	4.35×10^{-19}	2.11×10^{-19}	8.43×10^{-19}	1.26×10^{-18}
V_{xxy}	64800	0	2.81×10^{-19}	2.99×10^{-20}	5.67×10^{-20}	7.80×10^{-36}	1.40×10^{-19}	1.40×10^{-19}
V_{xxz}	64800	1.56×10^{-16}	1.63×10^{-16}	1.59×10^{-16}	1.59×10^{-16}	1.59×10^{-16}	1.60×10^{-16}	1.61×10^{-16}
V_{xyy}	64800	0	8.43×10^{-19}	9.56×10^{-20}	1.45×10^{-19}	7.02×10^{-20}	2.81×10^{-19}	4.21×10^{-19}
V_{xyz}	64800	0	2.18×10^{-19}	2.08×10^{-20}	4.29×10^{-20}	6.71×10^{-36}	1.09×10^{-19}	1.09×10^{-19}
V_{xzz}	64800	0	2.28×10^{-18}	2.78×10^{-19}	3.97×10^{-19}	2.18×10^{-19}	6.88×10^{-19}	1.25×10^{-18}
V_{yyy}	64800	0	8.43×10^{-19}	8.98×10^{-20}	1.70×10^{-19}	0	4.21×10^{-19}	4.21×10^{-19}
V_{yyz}	64800	1.56×10^{-16}	1.63×10^{-16}	1.59×10^{-16}	1.59×10^{-16}	1.59×10^{-16}	1.60×10^{-16}	1.61×10^{-16}
V_{yzz}	64800	0	1.12×10^{-18}	1.20×10^{-19}	2.06×10^{-19}	3.51×10^{-20}	3.44×10^{-19}	5.62×10^{-19}
V_{zzz}	64800	3.05×10^{-16}	3.30×10^{-16}	3.17×10^{-16}	3.17×10^{-16}	3.17×10^{-16}	3.20×10^{-16}	3.23×10^{-16}

Table S9: Absolute-residual statistics for the Laplace–Poisson identities in Test No. 9 over 160,801 points.

Identity	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
$V_{xx} + V_{yy} + V_{zz}$	160801	0	1.13×10^{-9}	3.44×10^{-12}	3.30×10^{-11}	4.25×10^{-15}	5.21×10^{-13}	6.97×10^{-11}
$V_{xxx} + V_{xyy} + V_{xzz}$	160801	0	1.48×10^{-10}	3.04×10^{-13}	4.35×10^{-12}	8.20×10^{-18}	2.98×10^{-15}	3.31×10^{-12}
$V_{xxy} + V_{yyy} + V_{yzz}$	160801	0	4.13×10^{-10}	6.26×10^{-13}	9.54×10^{-12}	1.02×10^{-17}	4.65×10^{-15}	3.74×10^{-12}
$V_{xxz} + V_{yyz} + V_{zzz}$	160801	0	8.10×10^{-10}	1.58×10^{-12}	1.96×10^{-11}	2.19×10^{-17}	1.32×10^{-14}	1.91×10^{-11}

Table S10: Absolute-residual statistics for the Laplace–Poisson identities in Test No. 10 over 160,801 points.

Identity	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
$V_{xx} + V_{yy} + V_{zz}$	160801	0	1.05×10^{-9}	3.42×10^{-12}	3.27×10^{-11}	3.80×10^{-15}	4.63×10^{-13}	7.43×10^{-11}
$V_{xxx} + V_{xyy} + V_{xzz}$	160801	0	1.58×10^{-10}	3.44×10^{-13}	4.77×10^{-12}	7.81×10^{-18}	3.29×10^{-15}	3.12×10^{-12}
$V_{xxy} + V_{yyy} + V_{yzz}$	160801	0	3.73×10^{-10}	5.90×10^{-13}	9.16×10^{-12}	9.41×10^{-18}	4.20×10^{-15}	3.48×10^{-12}
$V_{xxz} + V_{yyz} + V_{zzz}$	160801	0	7.29×10^{-10}	1.59×10^{-12}	1.92×10^{-11}	1.68×10^{-17}	1.14×10^{-14}	2.05×10^{-11}

Table S11: Absolute-residual statistics for the Laplace–Poisson identities in Test No. 11 over 160,801 points.

Identity	N	Minimum	Maximum	Mean	RMSE	P50	P90	P99
$V_{xx} + V_{yy} + V_{zz}$	160801	0	1.79×10^{-10}	5.65×10^{-13}	2.50×10^{-12}	3.91×10^{-14}	9.57×10^{-13}	1.01×10^{-11}
$V_{xxx} + V_{xyy} + V_{xzz}$	160801	0	8.10×10^{-11}	6.12×10^{-14}	9.20×10^{-13}	6.10×10^{-17}	5.77×10^{-15}	6.18×10^{-13}
$V_{xxy} + V_{yyy} + V_{yzz}$	160801	0	8.06×10^{-11}	5.57×10^{-14}	7.48×10^{-13}	7.09×10^{-17}	6.79×10^{-15}	7.30×10^{-13}
$V_{xxz} + V_{yyz} + V_{zzz}$	160801	0	2.91×10^{-14}	5.79×10^{-16}	1.96×10^{-15}	4.05×10^{-17}	1.15×10^{-15}	1.04×10^{-14}

5 Runtime audit of Test No. 12

The runtime audit is regenerated from the complete 401-by-401 timing grid. It retains all measurements, verifies the stored cumulative column against the sum of the 20 component columns, and separates all-point, inside-projection, and outside-projection distributions.

5.1 Statistical audit of Test No. 12

The tables are regenerated from all 160,801 archived timing records. P50, P90, and P99 are empirical percentiles; no timing sample is removed.

Table S12: Per-component single-core timing statistics for Test No. 12. Each point is the mean of ten timed evaluations after two warm-up evaluations.

Component	N	Minimum	Maximum	Mean	P50	P90	P99
V	160801	7.10×10^{-4}	2.32×10^{-1}	3.72×10^{-3}	2.27×10^{-3}	3.59×10^{-3}	1.08×10^{-1}
V_x	160801	1.40×10^{-3}	5.20×10^{-1}	6.44×10^{-3}	4.10×10^{-3}	6.31×10^{-3}	1.68×10^{-1}
V_y	160801	1.39×10^{-3}	3.67×10^{-1}	5.11×10^{-3}	4.13×10^{-3}	6.27×10^{-3}	9.25×10^{-3}
V_z	160801	1.39×10^{-3}	3.95×10^{-1}	8.16×10^{-3}	3.88×10^{-3}	6.13×10^{-3}	1.78×10^{-1}
V_{xx}	160801	2.04×10^{-3}	7.62×10^{-1}	4.94×10^{-2}	7.14×10^{-3}	2.47×10^{-1}	5.09×10^{-1}
V_{xy}	160801	2.64×10^{-3}	1.13×10^{0}	5.82×10^{-2}	9.74×10^{-3}	3.17×10^{-1}	6.34×10^{-1}
V_{yy}	160801	2.05×10^{-3}	1.02×10^{0}	3.31×10^{-2}	6.99×10^{-3}	1.28×10^{-2}	5.08×10^{-1}
V_{zx}	160801	2.58×10^{-3}	1.25×10^{0}	4.12×10^{-2}	8.38×10^{-3}	1.71×10^{-2}	3.52×10^{-1}
V_{zy}	160801	2.61×10^{-3}	1.35×10^{0}	5.89×10^{-2}	8.86×10^{-3}	3.17×10^{-1}	6.11×10^{-1}
V_{zz}	160801	2.02×10^{-3}	1.05×10^{0}	1.24×10^{-2}	6.01×10^{-3}	9.15×10^{-3}	2.62×10^{-1}
V_{xxx}	160801	2.86×10^{-3}	1.49×10^{0}	2.02×10^{-2}	1.10×10^{-2}	1.88×10^{-2}	7.25×10^{-1}
V_{xxy}	160801	4.31×10^{-3}	2.33×10^{0}	1.15×10^{-1}	1.71×10^{-2}	5.25×10^{-1}	1.09×10^{0}
V_{xxz}	160801	4.30×10^{-3}	2.32×10^{0}	1.08×10^{-1}	1.62×10^{-2}	5.22×10^{-1}	1.09×10^{0}
V_{xyz}	160801	5.86×10^{-3}	2.99×10^{0}	4.93×10^{-2}	2.10×10^{-2}	3.54×10^{-2}	1.46×10^{0}
V_{yyx}	160801	4.31×10^{-3}	2.33×10^{0}	9.51×10^{-2}	1.71×10^{-2}	5.18×10^{-1}	1.09×10^{0}
V_{yyy}	160801	2.93×10^{-3}	1.51×10^{0}	2.07×10^{-2}	1.13×10^{-2}	1.89×10^{-2}	7.35×10^{-1}
V_{yyz}	160801	4.33×10^{-3}	2.27×10^{0}	8.79×10^{-2}	1.62×10^{-2}	5.14×10^{-1}	1.08×10^{0}
V_{zzx}	160801	4.28×10^{-3}	2.15×10^{0}	2.80×10^{-2}	1.37×10^{-2}	2.14×10^{-2}	1.07×10^{0}
V_{zzy}	160801	4.23×10^{-3}	2.21×10^{0}	3.04×10^{-2}	1.40×10^{-2}	2.17×10^{-2}	1.08×10^{0}
V_{zzz}	160801	2.37×10^{-3}	1.50×10^{0}	1.66×10^{-2}	8.54×10^{-3}	1.32×10^{-2}	3.86×10^{-1}

Table S13: Derivative-rank and cumulative timing statistics for Test No. 12. The final column compares mean costs inside and outside the tesseroid projection.

Quantity	N	Minimum	Maximum	Mean	P50	P90	P99	Inside/outside mean
potential	160801	7.10×10^{-4}	2.32×10^{-1}	3.72×10^{-3}	2.27×10^{-3}	3.59×10^{-3}	1.08×10^{-1}	0.78
first-order sum	160801	4.99×10^{-3}	7.72×10^{-1}	1.97×10^{-2}	1.24×10^{-2}	1.91×10^{-2}	1.92×10^{-1}	0.89
second-order sum	160801	1.95×10^{-2}	5.41×10^{0}	2.53×10^{-1}	5.63×10^{-2}	6.33×10^{-1}	1.46×10^{0}	1.86
third-order sum	160801	5.75×10^{-2}	2.02×10^{1}	5.71×10^{-1}	1.46×10^{-1}	1.69×10^{0}	8.70×10^{0}	2.50
all 20 components	160801	8.58×10^{-2}	2.62×10^{1}	8.47×10^{-1}	2.94×10^{-1}	2.11×10^{0}	9.93×10^{0}	2.23